Note Title 2012/05/23

Unit 6.3 The dot product.

Section 12.3 p800 - 803.

<u>Definition</u> If $\bar{a} = \langle a_1, a_2 \rangle$ and $\bar{b} = \langle b_1, b_2 \rangle$

then the dot product of $\bar{a}$ and $\bar{b}$ is the <u>number</u>

$$\bar{a} \cdot \bar{b} = a_1 b_1 + a_2 b_2$$

in $\mathbb{R}^2$

If $\bar{a} = \langle a_1, a_2, a_3 \rangle$ and $\bar{b} = \langle b_1, b_2, b_3 \rangle$

Then

$\bar{a} \cdot \bar{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$

in $\mathbb{R}^3$

Examples

1. $\langle -2, 3 \rangle \cdot \langle 4, -5 \rangle$

2. $(8\bar{i} - 2\bar{j} + 8\bar{k}) \cdot (2\bar{i} + 4\bar{k})$

3. $\langle 2, 3, -1 \rangle \cdot \bar{j}$

Solutions:

1. $\langle -2, 3\rangle \cdot \langle 4, -5\rangle$

$= -2(4) + 3(-5) = -23$

2. $(8\bar{i} - 2\bar{j} + 3\bar{k}) \cdot (2\bar{i} + 0\bar{j} + 4\bar{k})$

$= \langle 8, -2, 3\rangle \cdot \langle 2, 0, 4\rangle$

$= 8(2) + (-2)(0) + 3(4) = 28$

3. $\langle 2, 3, -1\rangle \cdot \langle 0, 1, 0\rangle = 3$

Propertise of the Dot Product.

- Let $\bar{a}$, $\bar{b}$ and $\bar{c}$ be vectors in $\mathbb{R}^n$;
$n = 2, 3$ and $k \in \mathbb{R}$.

1. $\bar{a} \cdot \bar{a} = |\bar{a}|^2$

2. $\bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{a}$

3. $\bar{a} \cdot (\bar{b} + \bar{c}) = \bar{a} \cdot \bar{b} + \bar{c} \cdot \bar{a}$

4. $(k\bar{a}) \cdot \bar{b} = k(\bar{a} \cdot \bar{b})$ and
$\bar{a} \cdot (k\bar{b}) = k(\bar{a} \cdot \bar{b})$

5. $\bar{0} \cdot \bar{a} = 0$

Proof 1.

For $n=2$ Let $\bar{a} = \langle a_1, a_2 \rangle$

Then LHS

$$\bar{a} \cdot \bar{a} = \langle a_1, a_2 \rangle \cdot \langle a_1, a_2 \rangle$$

$$= (a_1)(a_1) + (a_2)(a_2) = a_1^2 + a_2^2$$

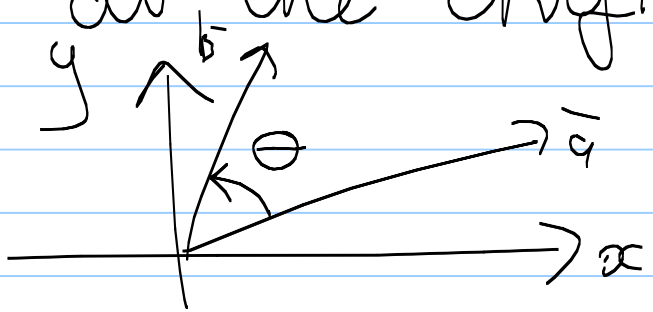
RHS $|\bar{a}|^2 = \left(\sqrt{a_1^2 + a_2^2}\right)^2$

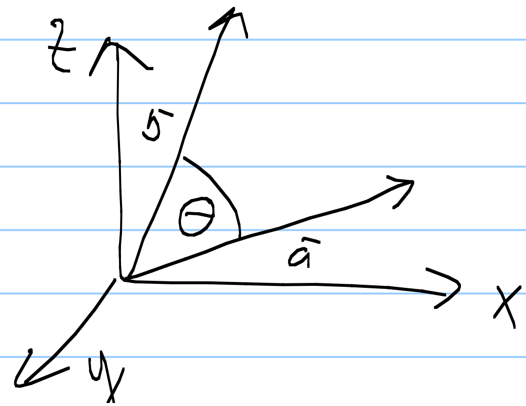
$$= a_1^2 + a_2^2$$

$\therefore$ LHS = RHS.

Geometric interpretation

The dot product $\bar{a}\cdot\bar{b}$ can be interpreted in terms of the angle θ $(0 \leq \theta \leq \pi)$ between $\bar{a}$ and $\bar{b}$. ($\bar{a}$ and $\bar{b}$ start at the origin)





Note: If $\bar{a}$ and $\bar{b}$ are parallel vectors, then $\theta = 0$ or $\theta = \pi$.

Theorem

If θ is the angle between the vectors $\bar{a}$ and $\bar{b}$, then

$$\bar{a} \cdot \bar{b} = |\bar{a}|\,|\bar{b}| \cos\theta .$$

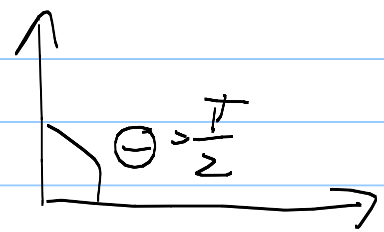
Corollary

If $\bar{a}$ and $\bar{b}$ are non-zero vectors then we write

$$\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$$

Definition

Two vectors $\bar{a}$ and $\bar{b}$ are called orthogonal (perpendicular) if the angle between the vectors is $\theta = \frac{\pi}{2}$



Definition

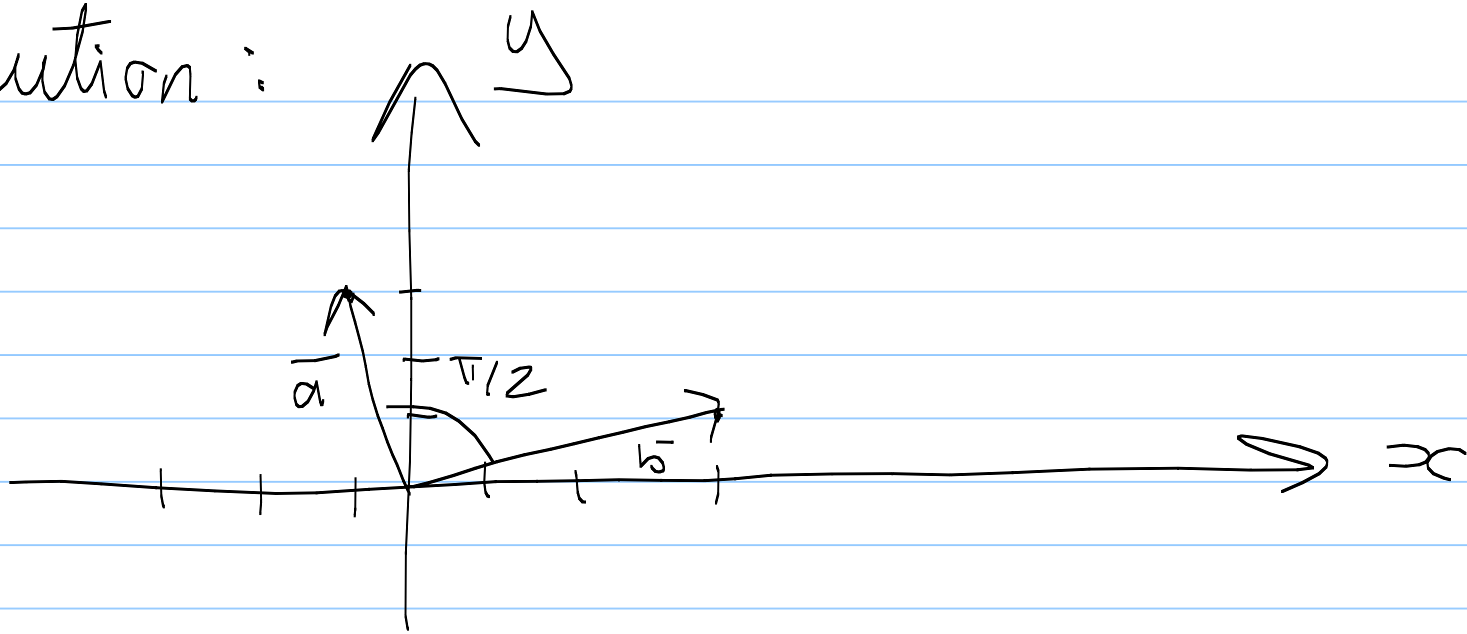
Two non-zero vectors $\bar{a}$ and $\bar{b}$ are orthogonal if and only if

$$\bar{a} \cdot \bar{b} = 0$$

Escamples

Are the vectors $\bar{a} = \langle -1, 3 \rangle$ and $\bar{b} = \langle 3, 1 \rangle$ orthogonal?

Solution :



$$\bar{a} \cdot \bar{b} = \langle -1, 3 \rangle \cdot \langle 3, 1 \rangle$$
$$= -3 + 3 = 0 .$$

Yes.

Example

Find k such that

$\bar{a} = \langle -1, 3, k \rangle$ and

$\bar{b} = \langle 2, k, 4 \rangle$ are orthogonal.

Solution :

$$\bar{a} \cdot \bar{b} = \langle -1, 3, K \rangle \cdot \langle 2, K, 4 \rangle$$

$$= -2 + 3K + 4K$$

$$= -2 + 7K$$

$\bar{a}$ and $\bar{b}$ are orThogonal iff

$$\bar{a} \cdot \bar{b} = 0$$

$$\Rightarrow \quad -2 + 7K = 0 \quad \Rightarrow K = 2/7$$